Introduction to Economics

Class 11 — Production and costs

Pavel Kocourek

Universitat de Barcelona · Group A6

29 October 2026

Quiz 8 — Graphs

20 minutes

Two sheets, both sides: Part 1 is the first sheet, Part 2 the second. Answer both.

Write each number in its box and the labels inside their areas on the graph; calculations in the Working space.

Write your name on both sheets.

At the end: photograph the code at the top left (your result appears there), then hand in *both sheets*.

Plan for today

- What economists count as a cost: explicit and implicit costs, opportunity cost
- From workers to output: the production function
- Total, average and marginal cost
- Why the cost curves are U-shaped
- Short run and long run; economies and diseconomies of scale

Warm-up: what does the concert cost you?

Tonight: a concert or your supermarket shift?

Ticket paid: 30, no refund · shift pays 45 · bus to the concert: 2

1. Your opportunity cost of going?

2. A friend offers 40 for the ticket. Now?

3. The concert is worth 60 to you. Go?

In pairs, two minutes; then hands per item.

Warm-up: what does the concert cost you?

Tonight: a concert or your supermarket shift?

Ticket paid: 30, no refund · shift pays 45 · bus to the concert: 2

1. Your opportunity cost of going?
   45 + 2 = 47 euros: the ticket is sunk
2. A friend offers 40 for the ticket. Now?
3. The concert is worth 60 to you. Go?

Warm-up: what does the concert cost you?

Tonight: a concert or your supermarket shift?

Ticket paid: 30, no refund · shift pays 45 · bus to the concert: 2

1. Your opportunity cost of going?
   45 + 2 = 47 euros: the ticket is sunk
2. A friend offers 40 for the ticket. Now?
   going costs 87: the 40 is given up too
3. The concert is worth 60 to you. Go?

Warm-up: what does the concert cost you?

Tonight: a concert or your supermarket shift?

Ticket paid: 30, no refund · shift pays 45 · bus to the concert: 2

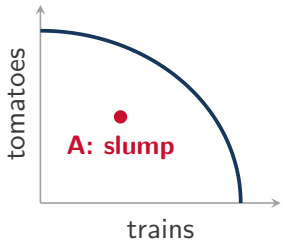
1. Your opportunity cost of going?
   $45 + 2 = 47$ euros: the ticket is sunk
2. A friend offers 40 for the ticket. Now?
   going costs 87: the 40 is given up too
3. The concert is worth 60 to you. Go?
   no offer: go ($60 > 47$); with the offer: sell and work ($60 < 87$)

Warm-up: what moves the frontier?

A country makes trains and tomatoes

1. A greenhouse method doubles the tomatoes per worker
2. Workers arrive who can do either job
3. Factories idle in a slump reopen

In pairs, one minute; then hands.

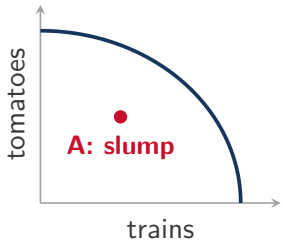


Warm-up: what moves the frontier?

A country makes trains and tomatoes

1. A greenhouse method doubles the tomatoes per worker

   the tomato end only: the frontier swings out
2. Workers arrive who can do either job
3. Factories idle in a slump reopen



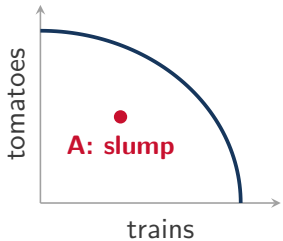
Warm-up: what moves the frontier?

A country makes trains and tomatoes

1. A greenhouse method doubles the tomatoes per worker

   the tomato end only: the frontier swings out
2. Workers arrive who can do either job

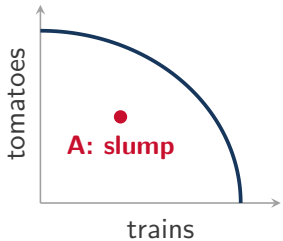
   both ends move out
3. Factories idle in a slump reopen



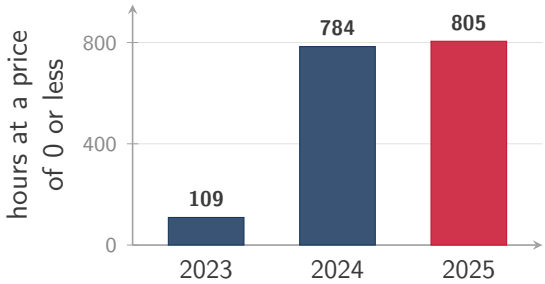
Warm-up: what moves the frontier?

A country makes trains and tomatoes

1. A greenhouse method doubles the tomatoes per worker
   the tomato end only: the frontier swings out
2. Workers arrive who can do either job
   both ends move out
3. Factories idle in a slump reopen
   a move toward the frontier, not a shift



Spain, 2025: about 800 hours of electricity at a price of zero or less



Almost all in the middle of the day, most in spring.

Why would anyone sell electricity for nothing?

Own count from the day-ahead prices of OMIE (the Spanish–Portuguese wholesale electricity market), mainland Spain (set per quarter-hour since October 2025: four of them count as one hour).

Counting what the owner gives up: the shop earns 1,700, not 4,400

The print shop's month

20 days · 4 printers · 220 bags a day at 4 euros

	euros
sales	17,600
− explicit costs (rent and press, printers)	13,200

Counting what the owner gives up: the shop earns 1,700, not 4,400

The print shop's month

20 days · 4 printers · 220 bags a day at 4 euros

	euros
sales	17,600
− explicit costs (rent and press, printers)	13,200
= accounting profit	**4,400**

Counting what the owner gives up: the shop earns 1,700, not 4,400

The print shop's month

20 days · 4 printers · 220 bags a day at 4 euros

	euros
sales	17,600
− explicit costs (rent and press, printers)	13,200
= accounting profit	**4,400**
− implicit costs (her pay at an agency, interest forgone)	2,700

Counting what the owner gives up: the shop earns 1,700, not 4,400

The print shop's month

20 days · 4 printers · 220 bags a day at 4 euros

	euros
sales	17,600
− explicit costs (rent and press, printers)	13,200
= accounting profit	**4,400**
− implicit costs (her pay at an agency, interest forgone)	2,700
= economic profit	**1,700**

Accountants count the money paid out; economists count every cost given up.

In pairs, three minutes: a yoga studio's profit

A yoga studio's year

sales	72,000
rent, teachers, heating	40,000

The owner left a job paying 29,000. Her 75,000 of savings in the studio could earn 4 % a year.

1. Accounting profit?
2. Implicit costs?
3. Economic profit?
4. Economic profit 0: is she losing money?

In pairs, three minutes: a yoga studio's profit

A yoga studio's year

sales	72,000
rent, teachers, heating	40,000

The owner left a job paying 29,000. Her 75,000 of savings in the studio could earn 4 % a year.

1. Accounting profit?
   72,000 − 40,000 = 32,000
2. Implicit costs?
3. Economic profit?
4. Economic profit 0: is she losing money?

In pairs, three minutes: a yoga studio's profit

A yoga studio's year

sales	72,000
rent, teachers, heating	40,000

The owner left a job paying 29,000. Her 75,000 of savings in the studio could earn 4 % a year.

1. Accounting profit?
   72,000 − 40,000 = 32,000
2. Implicit costs?
   29,000 + 3,000 = 32,000
3. Economic profit?
4. Economic profit 0: is she losing money?

In pairs, three minutes: a yoga studio's profit

A yoga studio's year

sales	72,000
rent, teachers, heating	40,000

The owner left a job paying 29,000. Her 75,000 of savings in the studio could earn 4 % a year.

1. Accounting profit?
   72,000 − 40,000 = 32,000
2. Implicit costs?
   29,000 + 3,000 = 32,000
3. Economic profit?
   0: sales just cover every cost
4. Economic profit 0: is she losing money?

In pairs, three minutes: a yoga studio's profit

A yoga studio's year

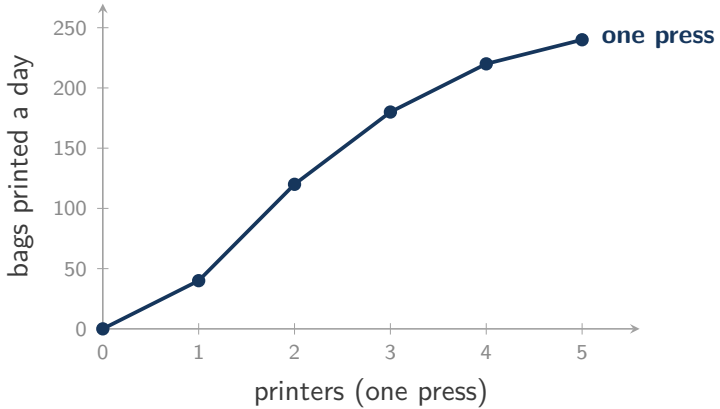
sales	72,000
rent, teachers, heating	40,000

The owner left a job paying 29,000. Her 75,000 of savings in the studio could earn 4 % a year.

1. Accounting profit?
   72,000 − 40,000 = 32,000
2. Implicit costs?
   29,000 + 3,000 = 32,000
3. Economic profit?
   0: sales just cover every cost
4. Economic profit 0: is she losing money?
   no: she earns what her best alternative pays

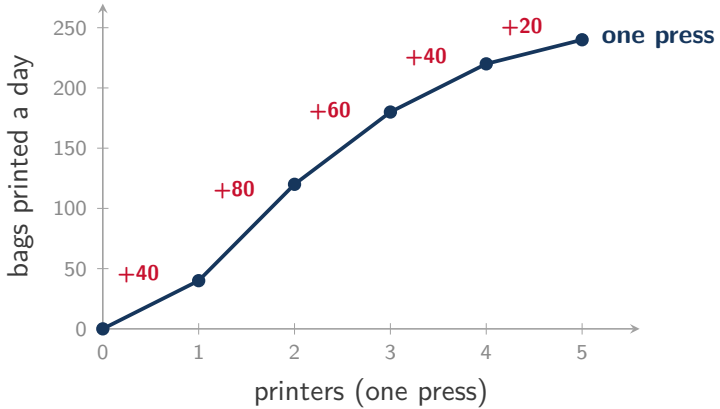
One press, more printers: from the third on, each adds fewer bags

A print shop in Poblenou: one press; customers bring the bags.



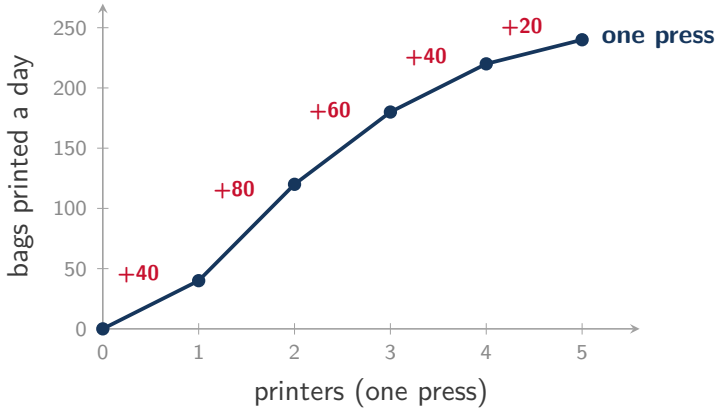
One press, more printers: from the third on, each adds fewer bags

A print shop in Poblenou: one press; customers bring the bags.



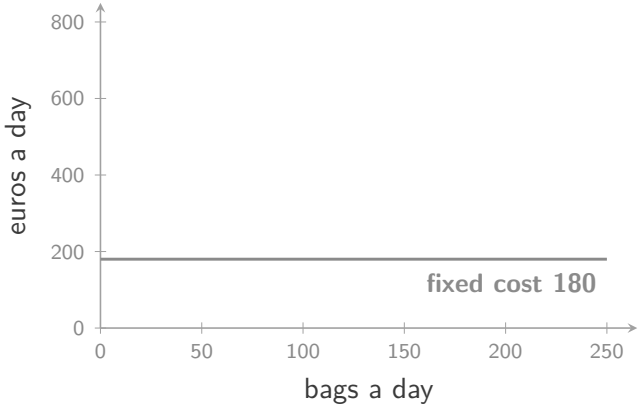
One press, more printers: from the third on, each adds fewer bags

A print shop in Poblenou: one press; customers bring the bags.

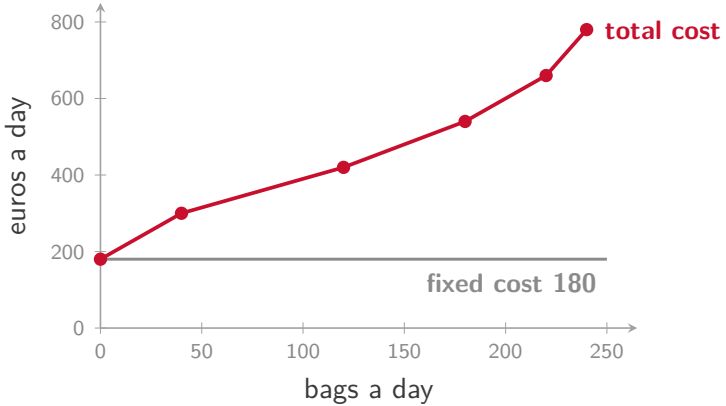


Marginal product: the bags one more printer adds. It falls once they crowd the press.

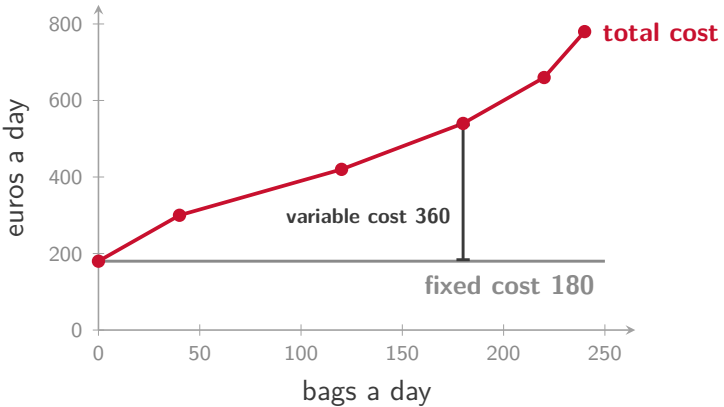
Total cost climbs faster as the press gets crowded



Total cost climbs faster as the press gets crowded



Total cost climbs faster as the press gets crowded



From the third printer on, each adds 120 but fewer bags: the curve gets steeper.

Per bag: the average bottoms out at 3.00; once the press is crowded, extra bags get dearer

printers	bags a day	total cost TC
0	0	180
1	40	300
2	120	420
3	180	540
4	220	660
5	240	780

Fixed cost 180 a day · each printer 120 a day.

Per bag: the average bottoms out at 3.00; once the press is crowded, extra bags get dearer

printers	bags a day	total cost TC	**AFC** FC ÷ bags	**AVC** VC ÷ bags	**ATC** TC ÷ bags
0	0	180	—	—	—
1	40	300	4.50	3.00	7.50
2	120	420	1.50	2.00	3.50
3	180	540	1.00	2.00	**3.00**
4	220	660	0.82	2.18	**3.00**
5	240	780	0.75	2.50	3.25

FC, VC: fixed, variable cost · AFC, AVC, ATC: average fixed, variable, total cost · MC: marginal cost

Fixed cost 180 a day · each printer 120 a day.

Per bag: the average bottoms out at 3.00; once the press is crowded, extra bags get dearer

printers	bags a day	total cost TC	**AFC** FC ÷ bags	**AVC** VC ÷ bags	**ATC** TC ÷ bags	**MC** extra cost ÷ extra bags
0	0	180	—	—	—	
1	40	300	4.50	3.00	7.50	3.00
2	120	420	1.50	2.00	3.50	1.50
3	180	540	1.00	2.00	**3.00**	2.00
4	220	660	0.82	2.18	**3.00**	3.00
5	240	780	0.75	2.50	3.25	6.00

FC, VC: fixed, variable cost · AFC, AVC, ATC: average fixed, variable, total cost · MC: marginal cost

Fixed cost 180 a day · each printer 120 a day.

MC = 120 ÷ the extra bags: it rises as each printer adds fewer.

In pairs, four minutes: an ice-cream maker's costs

An ice-cream maker's day

tubs a day	total cost
0	60
10	100
20	130
30	180
40	260

1. Its fixed cost?
2. ATC at 30 tubs?
3. MC of the tubs from 30 to 40?
4. The rent rises by 30 a day. ATC at 30? MC from 30 to 40?

In pairs, four minutes: an ice-cream maker's costs

An ice-cream maker's day

tubs a day	total cost
0	60
10	100
20	130
30	180
40	260

1. Its fixed cost?
   60 euros a day: the cost of making nothing
2. ATC at 30 tubs?
3. MC of the tubs from 30 to 40?
4. The rent rises by 30 a day. ATC at 30? MC from 30 to 40?

In pairs, four minutes: an ice-cream maker's costs

An ice-cream maker's day

tubs a day	total cost
0	60
10	100
20	130
30	180
40	260

1. Its fixed cost?
   60 euros a day: the cost of making nothing
2. ATC at 30 tubs?
   180 ÷ 30 = 6.00 euros a tub
3. MC of the tubs from 30 to 40?
4. The rent rises by 30 a day. ATC at 30? MC from 30 to 40?

In pairs, four minutes: an ice-cream maker's costs

An ice-cream maker's day

tubs a day	total cost
0	60
10	100
20	130
30	180
40	260

1. Its fixed cost?
   60 euros a day: the cost of making nothing
2. ATC at 30 tubs?
   180 ÷ 30 = 6.00 euros a tub
3. MC of the tubs from 30 to 40?
   80 ÷ 10 = 8.00 euros a tub
4. The rent rises by 30 a day. ATC at 30? MC from 30 to 40?

In pairs, four minutes: an ice-cream maker's costs

An ice-cream maker's day

tubs a day	total cost
0	60
10	100
20	130
30	180
40	260

1. Its fixed cost?
   60 euros a day: the cost of making nothing
2. ATC at 30 tubs?
   180 ÷ 30 = 6.00 euros a tub
3. MC of the tubs from 30 to 40?
   80 ÷ 10 = 8.00 euros a tub
4. The rent rises by 30 a day. ATC at 30? MC from 30 to 40?
   ATC 7.00; MC still 8.00

Three honey farms: how much does the market supply? (optional)

Jars a week each farm would sell

	at 6 euros	at 8 euros
Farm A	300	400
Farm B	0	250
Farm C	0	150

1. Market supply at 8 euros?
2. At 6 euros?
3. A fourth farm sells 100 at either price. At 6?

Three honey farms: how much does the market supply? (optional)

Jars a week each farm would sell

	at 6 euros	at 8 euros
Farm A	300	400
Farm B	0	250
Farm C	0	150

1. Market supply at 8 euros?
   800 jars: add the jars
2. At 6 euros?
3. A fourth farm sells 100 at either price. At 6?

Three honey farms: how much does the market supply? (optional)

Jars a week each farm would sell

	at 6 euros	at 8 euros
Farm A	300	400
Farm B	0	250
Farm C	0	150

1. Market supply at 8 euros?
   800 jars: add the jars
2. At 6 euros?
   300: only A sells
3. A fourth farm sells 100 at either price. At 6?

Three honey farms: how much does the market supply? (optional)

Jars a week each farm would sell

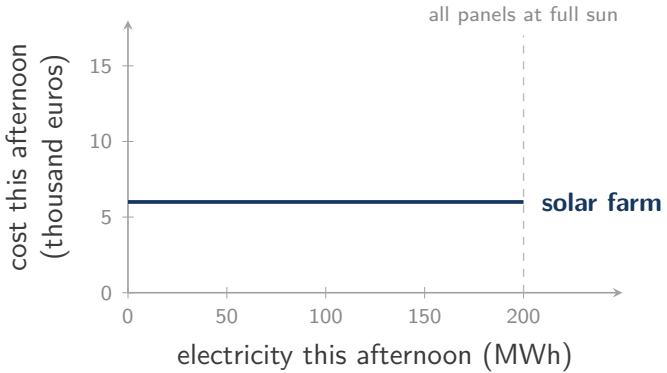
	at 6 euros	at 8 euros
Farm A	300	400
Farm B	0	250
Farm C	0	150

1. Market supply at 8 euros?
   800 jars: add the jars
2. At 6 euros?
   300: only A sells
3. A fourth farm sells 100 at either price. At 6?
   400 jars a week

Break

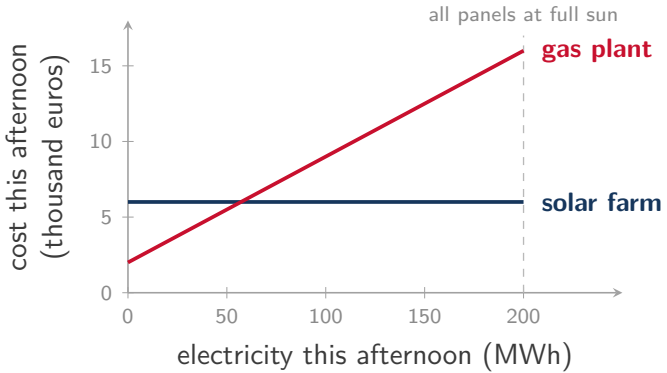
Back at 11:10

A solar farm's cost is almost all fixed: one more MWh costs about nothing



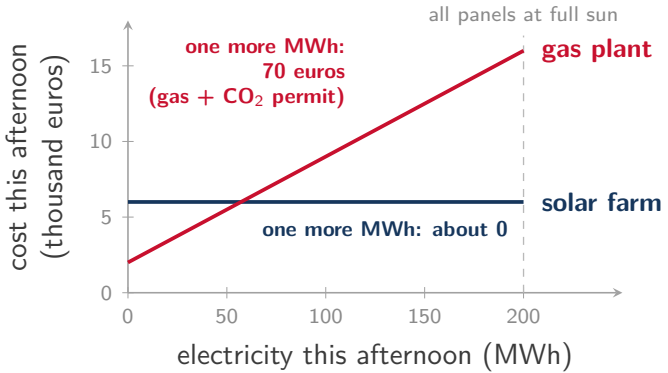
Illustrative numbers for one afternoon, not data.

A solar farm's cost is almost all fixed: one more MWh costs about nothing



Illustrative numbers for one afternoon, not data.

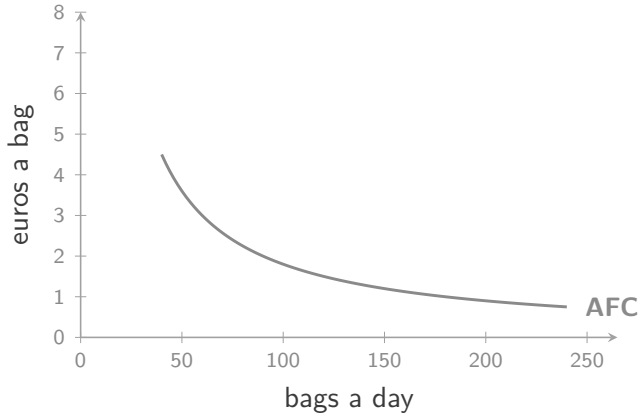
A solar farm's cost is almost all fixed: one more MWh costs about nothing



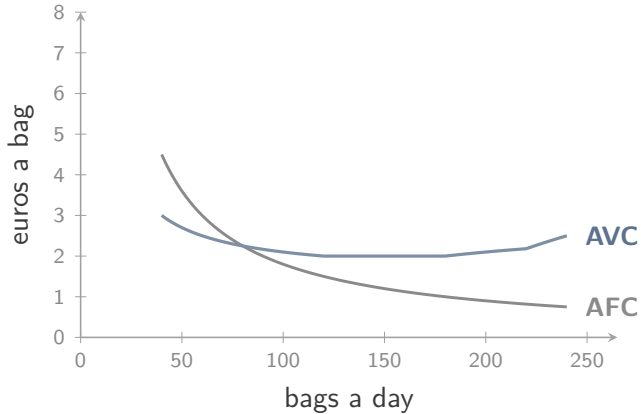
At a price of 0 the gas plant stops; switching off saves the solar farm nothing.

Illustrative numbers for one afternoon, not data.

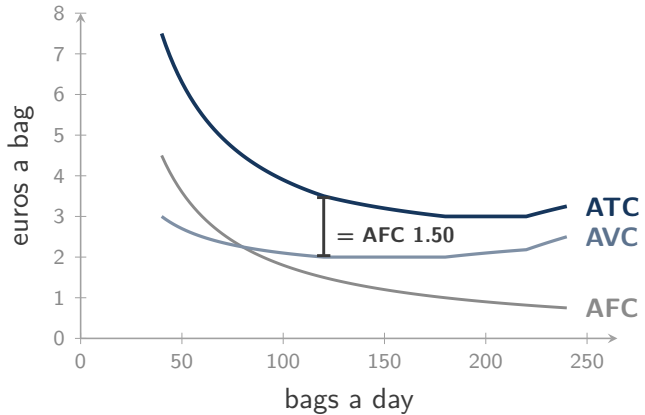
Fixed cost spread thinner, a crowded press: average total cost is a U



Fixed cost spread thinner, a crowded press: average total cost is a U

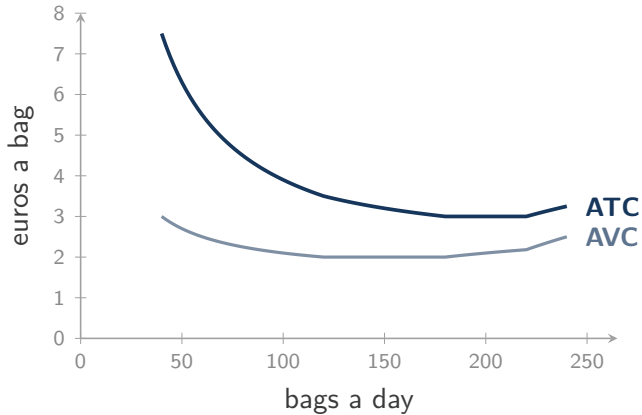


Fixed cost spread thinner, a crowded press: average total cost is a U

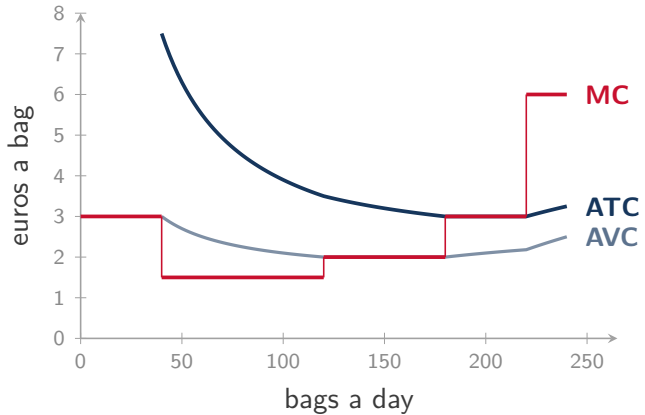


ATC = AFC + AVC: falling fixed cost per bag, then a crowded press.

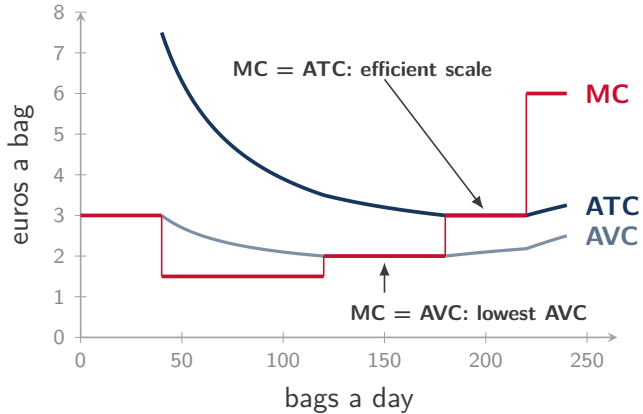
Marginal cost cuts through both averages at their lowest point



Marginal cost cuts through both averages at their lowest point



Marginal cost cuts through both averages at their lowest point



MC below the average pulls it down; above, pushes it up.

In pairs, two minutes: averages and the next one

Below the average or above it?

1. Average grade 6.0 after four exams; the fifth: 8.0. New average?
2. A firm's ATC is 6.00 at 100 units; the next 20 cost 3.00 each. ATC at 120?
3. Where MC equals ATC, ATC is...

Then hands per item.

In pairs, two minutes: averages and the next one

Below the average or above it?

1. Average grade 6.0 after four exams; the fifth: 8.0. New average?
   6.4: above the average, it rises
2. A firm's ATC is 6.00 at 100 units; the next 20 cost 3.00 each. ATC at 120?
3. Where MC equals ATC, ATC is...

In pairs, two minutes: averages and the next one

Below the average or above it?

1. Average grade 6.0 after four exams; the fifth: 8.0. New average?
   6.4: above the average, it rises
2. A firm's ATC is 6.00 at 100 units; the next 20 cost 3.00 each. ATC at 120?
   5.50: below the average, it falls
3. Where MC equals ATC, ATC is...

In pairs, two minutes: averages and the next one

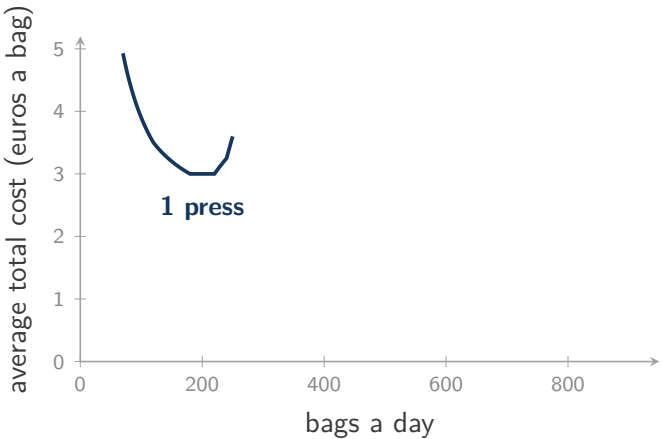
Below the average or above it?

1. Average grade 6.0 after four exams; the fifth: 8.0. New average?
   6.4: above the average, it rises
2. A firm's ATC is 6.00 at 100 units; the next 20 cost 3.00 each. ATC at 120?
   5.50: below the average, it falls
3. Where MC equals ATC, ATC is...
   at its bottom: neither pulled down nor up

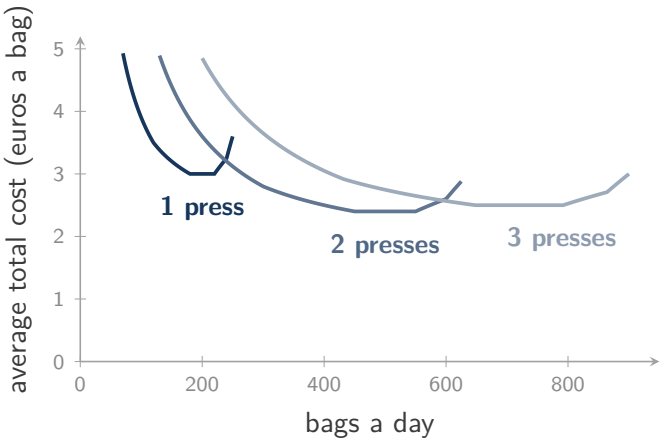
In the long run every cost can change: the shop can rent a second press

- Short run: the press is fixed; only printers change
- Long run: a second press, a bigger workshop
- In the long run every cost is variable
- How long? A print shop: months; a car plant: years

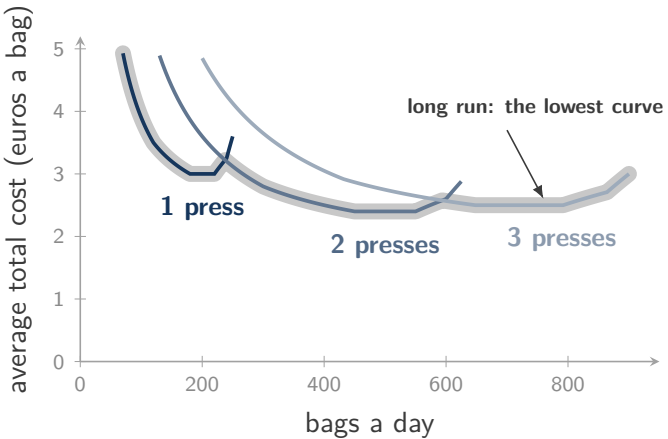
The long-run cost curve: the cheapest size for each output



The long-run cost curve: the cheapest size for each output



The long-run cost curve: the cheapest size for each output



In the long run the shop picks its size: the lowest curve at each output.

Double every input: if output more than doubles, the cost per bag falls

Now: 1 press, 4 printers: 220 bags for 660 a day, 3.00 a bag

Double every input: 2 presses, 8 printers, 1,320 a day.

if it makes	cost per bag

Double every input: if output more than doubles, the cost per bag falls

Now: 1 press, 4 printers: 220 bags for 660 a day, 3.00 a bag

Double every input: 2 presses, 8 printers, 1,320 a day.

if it makes	cost per bag	
550 bags	2.40	**economies of scale**

Double every input: if output more than doubles, the cost per bag falls

Now: 1 press, 4 printers: 220 bags for 660 a day, 3.00 a bag

Double every input: 2 presses, 8 printers, 1,320 a day.

if it makes	cost per bag	
550 bags	2.40	**economies of scale**
440 bags	3.00	**constant returns to scale**

Double every input: if output more than doubles, the cost per bag falls

Now: 1 press, 4 printers: 220 bags for 660 a day, 3.00 a bag

Double every input: 2 presses, 8 printers, 1,320 a day.

if it makes	cost per bag	
550 bags	2.40	**economies of scale**
440 bags	3.00	**constant returns to scale**
330 bags	4.00	**diseconomies of scale**

What counts: does output grow faster than cost?

Bigger is cheaper per bag — until coordination gets hard

Bigger, cheaper

workers who each do one task
bulk buying: ink by the barrel
machines that only come big
cheaper loans

Bigger is cheaper per bag — until coordination gets hard

Bigger, cheaper

workers who each do one task
bulk buying: ink by the barrel
machines that only come big
cheaper loans

Too big, dearer

more managers, slower decisions
messages lost on the way
workers far from the big picture

In pairs, five minutes: a laundry grows

A laundry for hotels New machines take months to arrive.

	machines	staff	kilos a day	cost a day
now	4	2	300	600
doubled	8	4	750	1,200
doubled again	16	8	1,200	2,400

1. ATC now?
2. Doubled: ATC? What kind of returns?
3. Doubled again: ATC? Returns?
4. A hotel adds 100 kilos a day from next week. What can change by then?

In pairs, five minutes: a laundry grows

A laundry for hotels New machines take months to arrive.

	machines	staff	kilos a day	cost a day
now	4	2	300	600
doubled	8	4	750	1,200
doubled again	16	8	1,200	2,400

1. ATC now?
   2.00 euros a kilo
2. Doubled: ATC?
   What kind of returns?
3. Doubled again: ATC? Returns?
4. A hotel adds 100 kilos a day from next week. What can change by then?

In pairs, five minutes: a laundry grows

A laundry for hotels New machines take months to arrive.

	machines	staff	kilos a day	cost a day
now	4	2	300	600
doubled	8	4	750	1,200
doubled again	16	8	1,200	2,400

1. ATC now?
   2.00 euros a kilo
2. Doubled: ATC?
   What kind of returns?
   1.60: economies of scale
3. Doubled again: ATC? Returns?
4. A hotel adds 100 kilos a day from next week. What can change by then?

In pairs, five minutes: a laundry grows

A laundry for hotels New machines take months to arrive.

	machines	staff	kilos a day	cost a day
now	4	2	300	600
doubled	8	4	750	1,200
doubled again	16	8	1,200	2,400

1. ATC now?
   2.00 euros a kilo
2. Doubled: ATC?
   What kind of returns?
   1.60: economies of scale
3. Doubled again: ATC? Returns?
   up to 2.00: diseconomies of scale
4. A hotel adds 100 kilos a day from next week. What can change by then?

In pairs, five minutes: a laundry grows

A laundry for hotels New machines take months to arrive.

	machines	staff	kilos a day	cost a day
now	4	2	300	600
doubled	8	4	750	1,200
doubled again	16	8	1,200	2,400

1. ATC now?
   2.00 euros a kilo
2. Doubled: ATC?
   What kind of returns?
   1.60: economies of scale
3. Doubled again: ATC? Returns?
   up to 2.00: diseconomies of scale
4. A hotel adds 100 kilos a day from next week. What can change by then?
   staff hours, not machines: the short run

Tote bags: which curve moves? (optional)

The market for printed tote bags

1. Printers' pay rises
2. The charge for a plastic bag doubles
3. Demand rises and totes get dearer: does supply shift too?

Tote bags: which curve moves? (optional)

The market for printed tote bags

1. Printers' pay rises
   supply of totes shifts left
2. The charge for a plastic bag doubles
3. Demand rises and totes get dearer: does supply shift too?

Tote bags: which curve moves? (optional)

The market for printed tote bags

1. Printers' pay rises
   supply of totes shifts left
2. The charge for a plastic bag doubles
   demand for totes shifts right
3. Demand rises and totes get dearer: does supply shift too?

Tote bags: which curve moves? (optional)

The market for printed tote bags

1. Printers' pay rises
   supply of totes shifts left
2. The charge for a plastic bag doubles
   demand for totes shifts right
3. Demand rises and totes get dearer: does supply shift too?
   no: they print more, along supply

Cheap solar: positive or normative? (optional)

Cheap solar: three statements

1. "More solar panels bring more hours at a price of zero."

2. "Electricity should be cheap for everyone."

3. A and B want cheap electricity. B: "Slow down: zero prices will scare off investors."
   B's reason: values or a prediction?

Cheap solar: positive or normative? (optional)

Cheap solar: three statements

1. "More solar panels bring more hours at a price of zero."
   positive: data can settle it
2. "Electricity should be cheap for everyone."
3. A and B want cheap electricity. B: "Slow down: zero prices will scare off investors."
   B's reason: values or a prediction?

Cheap solar: three statements

1. "More solar panels bring more hours at a price of zero."
   positive: data can settle it
2. "Electricity should be cheap for everyone."
   normative: a judgement of values
3. A and B want cheap electricity. B: "Slow down: zero prices will scare off investors."
   B's reason: values or a prediction?

Cheap solar: positive or normative? (optional)

Cheap solar: three statements

1. "More solar panels bring more hours at a price of zero."
   positive: data can settle it
2. "Electricity should be cheap for everyone."
   normative: a judgement of values
3. A and B want cheap electricity. B: "Slow down: zero prices will scare off investors."
   B's reason: values or a prediction?
   a prediction: how investors respond

A cost is what the firm gives up, paid out or not.

Once workers crowd the press, each adds less,

so marginal cost rises; it cuts average cost at its lowest point.

In the long run every cost can change: economies of scale, then diseconomies.

Monday: how much a firm sells at a given price.

Before Monday 2 November

- Monday: perfect competition; monopoly begins
- Read Mankiw & Taylor: Ch. 5, pp. 123–137
- Problem set 2 (Block 3): your own version is on Campus Virtual; due Thursday 5 November, on paper
- Problem set 3 (Block 4): handed out Monday; due Thursday 12 November
- Practise in Sextant: the units the course page lists



pkocourek@ub.edu · pkocourek.com/teaching/intro-economics · pkocourek.com/sextant/?c=intro-econ